



Daily Drilling Report

Well ID: Forge 78B-32

Field: UTAHFORGE

Geothermal Resource Group, Inc.

Well Name: 78B-32

Sect: 32 Town: 26 Rng: 9W County: Beaver State: UT

Report No: 1

Report For 27-Jun-21

Operator:	University of Utah	Rig:	Frontier Rig 16	Spud Date:	28-Jun-21	Daily Cost / Mud (\$):	---
Measured Depth (ft):	128	Last Casing:		Wellbore:	Original Wellbore	AFE No.	AFE (\$)
Vertical Depth (ft):		Next Casing:	24.000 at 129	RKB Elevation (ft):	30.40	---	Actual (\$)
Proposed TD (ft):	9500	Last BOP Test:		Job Reference RKB (ft):		---	---
Hole Made (ft) / Hrs:	0 / 0.0	Next BOP Test:		Working Interest:		Totals:	---
Average ROP (ft/hr):						Well Cost (\$):	---
Drilling Days (act./plan):	1/21	Flat Days (act./plan):	0/9	Total Days (act./plan):	1/30	Days On Location:	1
Pers/Hrs:	Operator: 3 / 36	Contractor:	12 / 144	Service:	8 / 96	Other:	1 / 12
						Total:	24 / 288

Safety Summary: No incidents or events reported. 0 days since LTI. Conducted Safety Meeting.

Current Operations: Finished drilling 22 in hole to 421 ft
Circulating hole clean at 421 ft.

Planned Operations: POH and run 16 in casing and cement in place.
Nipple up and test 20 in BOP.

Toolpusher: Steve Caldwell , Justin Bristol

Wellsite Supervisors: Virgil Welch , Brian Gresham

Tel No.: 7132807438

Operations Summary

From	To	Elapsed	End MD(ft)	Code	Operations Description	Non-Prod
0:00	6:00	6.00	128	WOD	Wait on daylight.	
6:00	9:00	3.00	128	RIGU	Pre-Spud Inspection.	
9:00	16:00	7.00	128	1-01	Cut and removed conductor pipe at 4" above ground level. Rigged up cellar pump. Set pipe racks in place. Strapped 105 joints of 5" drill pipe 9 joints of 5" HWDP and readied for pick up. Mixed spud mud.	
16:00	17:00	1.00	128	10-341	Pick up and make up 5 stands of 5" HWDP.	
17:00	22:30	5.50	128	10-341	Pick up 114 joints of 5" drill pipe and rack back in derrick.	
22:30	0:00	1.50	128	3-34-3	Make up 22" surface hole assembly.	

Management Summary

Waited on Day Light and conducted Pre-Spud Rig Inspection.
 Strapped, OD, ID and Fishing Neck of HWDP.
 Picked up 5 stands of HWDP and stood back in Derrick.
 Picked up 114 joints (38 Stands) of 5" DP and stood back in derrick.
 Cleaned and strapped 16 in Casing.
 Made up Directional tools.

Comments

Rig on "Day Rate" at 0900 hours of June 27, 2021.

Note: Spoke to Jim Goddard with Department of Water Resources. Jim had received our start casing, cementing, and BOP testing schedule and waived witnessing of the 20 in BOP test provided testing to be done by a 3rd party. Results to be documented and sent to DWR.
Fuel on hand 7,637.

Activities prior to Rig Move/ Spud

June 8, 2021

- Drill 36" hole from surface to 100'. All material was alluvium (decomposed granite). Experienced no hole cave or sluff.
- weld lifting eyes on 24", move all dirt out of the way and prep to run conductor and cement same in the morning

June 9, 2021

- Check hole for fill / clean out to 100' / welding 3 joints of 24", 1/2" wall. 125.5#/ft conductor casing / set on bottom / pull rig forward / center and level conductor (102' total)
- From surface pour 14 cu. Yds (2 trucks) in annulus. Cement came to within 33' of surface.
- Cement is 4000 psi mix with 3/4" rock
- Prep 14" casing for mouse hole. Will top off conductor cement @ 8:00 tomorrow morning prior to drilling service hole. Crew released for the day. No accidents, incidents or injuries

June 10, 2021

- Dump 6 cu. Yds of 4000 psi w/ 3/4" rock ready mix around conductor bringing the level from 33'bgl to 5'bgl. Used another 2 cu yds to cement tower foundation on 78-32
- Let cement around conductor thicken
- Drill 26" mouse hole to 75' bgl
- Weld 3 each 25' long 14" line pipe pieces for the mouse hole and lower in the ground
- Cement the mouse shuck with 7 cu. Yds of 4000 psi mix w/ 3/4" rock. Level 5' bgl
- Using excavator dig hole for cellar and install 10'OD x 5' deep cellar ring
- Pour 1 foot of 4000 psi, w/ 3/4" rock in bottom of cellar

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Report No: 1

Report For 27-Jun-21

- Clean up site and release Wyoming Casing

June 21, 2021

- With H&B move company man quarters, tool pusher quarters, septic system, fresh water system, power distribution panel and generator from 56-32 to 78B-32 and rig up same. Move Command center from 16A to 78B-32 and rig up same

June 22, 2021

- MSE training with all 4 Frontier crews, and 3 pushers, 3 GRG supervisors, 2 GRG office staff, 1 mud man, 2 mud loggers, 2 solids control

June 23, 2021

- MSE training with all contractors

June 24, 2021

- Tear down and move everything off 56-32 location except for subbase. On 78B-32 set and rig up pump package and mud pits, solids equipment, SCR and motor package, derrick, tool house. Removed monkey board from derrick and welder cut all racking fingers out of board and re-install to accept 5.5" drill pipe

June 25, 2021

- Hold JSA with all crews involved with rig move, On 56-32 rig down subbase and load out same, clean location and call for dumpster pick up, 56-32 completed. On 78B-32 set matting boards and build subbase, install drawworks, hang derrick and string blocks, welder completed modifying racking board to accommodate 5.5" drill pie, install modified racking board and raise A-Legs, set mud loggers trailer, set solids equipment, unload 16", set 2 mud coolers, received 1st load of diesel, finish setting top dog houses and hanging wind walls, rig up Pason NOTE: Spoke to Jim Goddard with Department of Water Resources Jim had received messages of our start casing, cementing and BOP testing schedule and waived witnessing of the 20" BOP test provided testing be done by a 3rd party and results be documented and sent to DWR

June 26, 2021

- Raise derrick and unbridal

- Rig up Top drive, mud coolers, fill tanks w/ water prepare to PUDP

Bit/BHA/Workstring Information

					Depth	This Run		R.O.P.				Mud			Pump				
No	Run	Make	Model	Diam	In	Dist	Hrs	Avg	Max	WOB	RPM	Torque	Wt	Flow	Press	J. Vel	P. Drp	HPH	JIF
1	1	Smith	XR-C	22.000	128														
Jets: 28 28 28					Out: 421		Grade: Cutter: /			Dull: /		Wear:		Brgs:	Gge:	Pull:			
BHA - No. 1 - BIT, BS, XO, MWD, XO, MONEL, FLOAT, OTHER, XO, HWDP = 314.30																			

Rig Information

Equipment Problems:

Location Condition:

Transport:

Solids Control Information

Screen Sizes:	Top	Middle 1	Middle 2	Bottom	Equipment Usage (Hrs):	
Shaker No 1:	60	60	60	60		
Shaker No 2:	60	60	60	60	Centrifuge 1: 4	Centrifuge 2: 4
Shaker No 3:	60	60	60	60		

Bulk Inventory

Item Type	Units	Beginning	Used	Received	On Hand	Net
Diesel (OBM)		1,000	735	7,372	7,637	6,637

Drill Pipe Inventory

DP Size	Joints	Weight	Grade	Thread	DP Size	Joints	Weight	Grade	Thread
5"	114	19.5	S-135	4.5IF					

Safety Information

Meetings/Drills	Time	Description	
Safety	60	Mixing chemicals, pinch points, running casing / BHA	
First Aid Treatments:	0	Medical Treatments:	0
Lost Time Incidents:	0	Days Since LTI:	0
☐ BOP Test		☐ Crownamatic Check	



Daily Drilling Report

Well ID: Forge 78B-32

Field: UTAHFORGE

Geothermal Resource Group, Inc.

Well Name: 78B-32

Sect: 32 Town: 26 Rng: 9W County: Beaver State: UT

Report No: 1

Report For 27-Jun-21

Weather Information

Sky Condition:	Clear	Visibility:	10
Air Temperature:	56 degF	Bar. Pressure:	29.87
Wind Speed/Dir:	10 / SE	Wind Gusts:	12



Daily Drilling Report

Well ID: Forge 78B-32

Field: UTAHFORGE

Geothermal Resource Group, Inc.

Well Name: 78B-32

Sect: 32 Town: 26 Rng: 9W County: Beaver State: UT

Report No: 2

Report For 28-Jun-21

Operator:	University of Utah	Rig:	Frontier Rig 16	Spud Date:	28-Jun-21	Daily Cost / Mud (\$):	---
Measured Depth (ft):	421	Last Casing:	16.000 at 416	Wellbore:	Original Wellbore	AFE No.	---
Vertical Depth (ft):		Next Casing:	11.750 at 3,300	RKB Elevation (ft):	30.40	AFE (\$)	Actual (\$)
Proposed TD (ft):	9500	Last BOP Test:	28-Jun-21	Job Reference RKB (ft):		---	---
Hole Made (ft) / Hrs:	293 / 4.0	Next BOP Test:	19-Jul-21	Working Interest:		---	---
Average ROP (ft/hr):	73.25					Totals:	---
Drilling Days (act./plan):	2/21	Flat Days (act./plan):	0/9	Total Days (act./plan):	2/30	Well Cost (\$):	---
Pers/Hrs: Operator:	3 / 36	Contractor:	14 / 168	Service:	4 / 48	Other:	3 / 36
						Total:	24 / 288

Safety Summary: No incidents or events reported. Conducted BOP Test, Crown Check, Safety Meeting.

Current Operations: Testing out choke manifold at report time.

Planned Operations: Test casing along with BOPE.
Pick up 8" drill collars.
Trip in the hole with tri-cone bit.
Drill out shoe track and 10 ft of new formation to perform LOT
Pull out of the hole and lay down tricone bit
Make up 14 3/4" BHA with PDC and continue drilling 14 3/4" section.

Toolpusher: Steve Caldwell , Justin Bristol

Wellsite Supervisors: Virgil Welch , Brian Gresham

Tel No.: 7132807438

Operations Summary

From	To	Elapsed	End MD(ft)	Code	Operations Description	Non-Prod
0:00	1:00	1.00	128	3-34-3	Make up 22" surface hole assembly.	
1:00	1:30	0.50	128	03-051	Filled pipe and hole.	
1:30	5:00	3.50	421	03-021	Drilled 22" hole from 128' to 421'.	
5:00	5:30	0.50	421	03-051	Circulated well bore clean.	
5:30	6:00	0.50	421	3-27	Wiped hole f/ 421' t/ 100' with no issues. No fill on bottom.	
6:00	6:30	0.50	421	3-6-2	Pull out of the hole f/ 421'	
6:30	7:00	0.50	421	3-34-4	Lay down NMDC and bit	
7:00	8:00	1.00	421	4-56	Rig up casing crew.	
8:00	10:00	2.00	421	4-12-1	Rigged up and Ran 9 joints of 16" k-55 64# BTC casing (382.89') of total casing. Landed out with 33' landing joint. Shoe set at 416' KB. Float collar at 372' KB.	
10:00	11:00	1.00	421	4-56	Rig down casing crew	
11:00	12:30	1.50	421	4-12-1	Land casing with well head 3' below ground level.	
12:30	13:00	0.50	421	5-56	RIH with stab-in and stab into float collar at 378	
13:00	14:30	1.50	421	5-56	Rig up cement crew	
14:30	16:00	1.50	421	5-99	Pressure test to 1500 psi, Pump 20 bbl fresh water, pump 50 bbls of sepiolite, pump 5 bbl of fresh water, pump 13 bbl od sodium silkicate, pump 5 bbls fresh water, Mix and pump 146 bbl of 13.88 RC Theremelite w/ 2% CC, displace w/ 6 bbls fresh water, check floats CIP @ 16:00	
16:00	16:30	0.50	421	5-56	Rig down cement crew	
16:30	17:00	0.50	421	5-6-2	POH with stab-in	
17:00	0:00	7.00	421	1-14	Installed BOP as followed. 16" x 20" 2M DSA, 20" 2M spacer spool, 20" 2M x 20" 3M DSA, 20" 3M mud cross, 20" 3M X 20" 2M DSA, 20" 2M double gates, 20" 2M annular, 20" 2M flow spool.	

Management Summary

Drilled 22" hole from 128' to 421'. Circulated hole clean. Wiped hole from 421' to 100'.
Pulled out laying down tools.
Rigged up and ran 9 joints of 65 ppf, K-55 16" Buttress casing setting shoe at 416' KB.
Ran in the hole with stab-in tool.
Rigged up Resource cementing and cemented casing in place CIP @ 16:00.
Rigged down Resource cementing.
Pulled out of the hole with stab-in tool.
Installed 20" BOPE. Nipped up BOPE.

Comments

Good cement to surface.

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Field: UTAHFORGE

Geothermal Resource Group, Inc.

Well Name: 78B-32

Sect: 32 Town: 26 Rng: 9W County: Beaver State: UT

Report No: 2

Report For 28-Jun-21

Casing/Tubular Information

Type	Size (ins)	Top MD (ft)	Top TVD (ft)	Bottom MD (ft)	Bottom TVD (ft)	Hole Section	OH Diam. (ins)	Nom. Wgt. (lbs)	Nominal Grade	LOT (lbs/gal)
FULL	16.000	0		416		SURF	22.000	65	K-55	

Mud Information

																%			Gels			Temp		Mud
Dens.	Vis	PV	YP	Filt.	Cake	pH/ES	Solids	Oil	Water	Sand	LGS	Cl	Ca	CaCl	10s	10m	30m	In	Out	Loss				
28-Jun-21 05:15 at Depth 421 ft Mud Pits																								
8.90	51	19	22	12.2	1	9	11	0	89	2.5		800	120	0	9	16	25	85	85	10.0				

Bit/BHA/Workstring Information

				Depth	This Run		R.O.P.							Mud	Pump					
No	Run	Make	Model	Diam	In	Dist	Hrs	Avg	Max	WOB	RPM	Torque	Wt	Flow	Press	J. Vel	P. Drp	HHP	JIF	
1	1	Smith	XR-C	22.000	128	293	3.5	83.7	250.0	15	90	7	9	800	576	142	161	75	524	
Jets: 28 28 28					Out: 421		Grade: Cutter: 1 / 1		Dull: NO / NO				Wear: A		Brgs: 0		Gge: 0		Pull: TD	
BHA - No. 1 - BIT, BS, XO, MWD, XO, MONEL, FLOAT, OTHER, XO, HWDP = 314.30																				

Drilling Parameters

Depth (ft)		ROP (ft/hr)		WOB (lbs)		RPM		Torque (ft lbs)		Flow (gals/min)		Pressure (psi)	
From	To	Avg	Max	Avg	Max	Avg	Max	Avg	Max	Avg	Max		
128	421	80.0	120.0	15	30	75	80	3	8	650	800	1,100	
Annular Velocity: Drill Collars:				37.9	Drill Pipe:				34.7				

Miscellaneous Drilling Parameters

Hook Loads (lbs):	Off Bottom Rotate:	85	Pick Up:	100	Slack Off:	78	Drag Avg/Max:	3 / 5
Hours on BHA:	Since Inspection:	3.5	Total:		Jars:			

Survey Information

Survey Type		Meas. Depth	Inc.	Azimuth	TVD	Closure	Vertical Section	Coordinates N-S E-W		D.L.S.
MWD		241.0	0.69	358.3	241.0	1.5	1.5	N 1.5	W 0.0	0.286
MWD		337.0	1.04	359.5	337.0	2.9	2.9	N 2.9	W 0.1	0.365

Rig Information

Equipment Problems:	None
Location Condition:	Good.
Transport:	

Solids Control Information

Screen Sizes:	Top	Middle 1	Middle 2	Bottom	Equipment Usage (Hrs):			
Shaker No 1:	60	60	60	60	Desander: 4	Desilter: 4	Degasser: 0	
Shaker No 2:	60	60	60	60	Centrifuge 1: 4 (Solids Removal)			
Shaker No 3:	60	60	60	60				

Bulk Inventory

Item Type	Units	Beginning	Used	Received	On Hand	Net
Diesel (OBM)		7,637	940		6,697	

Drill Pipe Inventory

DP Size	Joints	Weight	Grade	Thread	DP Size	Joints	Weight	Grade	Thread
5"	114	19.5	S-135	4.5IF					

Safety Information

Meetings/Drills		Time	Description					
Safety		60	Running casing, cementing casing, Nipple up, Pinch points					
First Aid Treatments:		0	Medical Treatments:		0	Lost Time Incidents:	0	Days Since LTI:
☒ BOP Test		☒ Crownamatic Check						



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Well Name: 78B-32

Sect: 32 Town: 26 Rng: 9W County: Beaver State: UT

Report No: 2

Report For 28-Jun-21

Weather Information

Sky Condition:	Clear	Visibility:	10
Air Temperature:	66 degF	Bar. Pressure:	29.96
Wind Speed/Dir:	8 / NNE	Wind Gusts:	2

**Daily Drilling Report**

Well ID: Forge 78B-32

Field: UTAHFORGE

Geothermal Resource Group, Inc.

Well Name: 78B-32

Sect: 32 Town: 26 Rng: 9W County: Beaver State: UT

Report No: 3

Report For 29-Jun-21

Operator:	University of Utah	Rig:	Frontier Rig 16	Spud Date:	28-Jun-21	Daily Cost / Mud (\$):			---
Measured Depth (ft):	433	Last Casing:	16.000 at 416	Wellbore:	Original Wellbore	AFE No.	AFE (\$)	Actual (\$)	
Vertical Depth (ft):	433	Next Casing:	11.750 at 3,300	RKB Elevation (ft):	30.40	---	---	---	
Proposed TD (ft):	9500	Last BOP Test:	29-Jun-21	Job Reference RKB (ft):		---	---	---	
Hole Made (ft) / Hrs:	12 / 1.5	Next BOP Test:	19-Jul-21	Working Interest:		Totals:	---	---	
Average ROP (ft/hr):	8.0			LOT (lbs/gal):	19.20	Well Cost (\$):			---
Drilling Days (act./plan):	3/21	Flat Days (act./plan):	0/9	Total Days (act./plan):	3/30	Days On Location:			3
Pers/Hrs:	Operator: 3 / 36	Contractor:	14 / 168	Service:	4 / 48	Other:	3 / 36	Total:	24 / 288

Safety Summary: No incidents or events reported. Conducted BOP Test, Crown Check, Safety Meeting.**Current Operations:** Picked up RSS and ran in the hole. Drilled 14 3/4" hole section from 433' to 1,524' (report time)**Planned Operations:** Drill 14 3/4" section.**Toolpusher:** Steve Caldwell , Justin Bristol**Wellsite Supervisors:** Virgil Welch , Brian Gresham

Tel No.: 7132807438

Operations Summary

From	To	Elapsed	End MD(ft)	Code	Operations Description	Non-Prod
0:00	3:00	3.00	421	3-15	Installed inner and outer choke and kill valves. Installed choke and kill lines. Torqued all flanges.	
3:00	6:00	3.00	421	3-15	Attempted to Test BOPE system to 300 psi low to 1,000 psi high. First Low test of 300 psi testing against the pipe rams held. Stepped pressure up to 1,000 psi and pressure bleed back down to 500 psi. Attempted to pressure back up to 1,000 psi, pressure bleed back down to 500 psi. Inspected all surface equipment for leaks. No leaks were found. Decision was made to test choke manifold while allowing more time for cement to set. Tested out all valves on choke manifold to 300 psi low for 5 mins and 1,000 psi high for 10 mins.	
6:00	9:00	3.00	421	2-14	Nipple up Flow nipple w/ rotating head to flow line	
9:00	10:00	1.00	421	2-14	Walk catwalk into place and raise V-Door	
10:00	11:30	1.50	421	3-34-3	Set racks and move and strap 15 each 8" DCS	
11:30	14:30	3.00	421	3-34-3	Pick up 15 each 8" DCS and stand back in Derrick	
14:30	16:30	2.00	421	3-15	Test Blind rams to 250 psi for 5 min low & 1000 psi for 10 min high / test good / PU stand of drill pipe and test pipe rams to 250 psi for 5 min & 1000 psi for 10 min test good, test bag to 250 psi for 5 min low & 1000 psi high for 10 min	
16:30	18:00	1.50	421	3-34-3	PU 14-3/4" bit, bit sub, 3 stands of DCS and 1 stand of HWDP and RIH to drill float, cement and shoe	
18:00	18:30	0.50	421	2-14	Installed rotating rubber bushing to bearing assembly. Installed rotating rubber.	
18:30	19:30	1.00	421	3-28	Drilled out shoe track from 370' to 416'. Good cement Throughout shoe track.	
19:30	20:00	0.50	421	SERV	Serviced rig.	
20:00	20:45	0.75	421	3-28	Cleaned out rat hole to 421'. Drilled new formation to 433'	
20:45	21:00	0.25	421	3-5-1	Circulated hole clean for FIT.	
21:00	22:00	1.00	421	3-46	Performed FIT.	
22:00	23:30	1.50	421	3-6-2	Pulled out of the hole from 433' to surface. Broke out bit along with bit sub.	
23:30	0:00	0.50	421	3-98	Held detailed safety meeting with all involved personal on picking up BHA.	

Management Summary

Installed inner and outer choke and kill valves.
 Torqued up all breaks on BOPE.
 Attempted to test BOPE. High pressure test leaked off. Decision was made to allow more time for cement to set.
 Tested out choke manifold.
 Nipped up riser and installed flowline.
 Walked catwalk into place and raised V-Door.
 Set racks in place. Strapped 15-8" drill collars. Picked up 5 stands of 8" drill collars and racked back in the derrick.
 Tested out BOPE. Good test.
 Made up 14 3/4" slick assembly.
 Tripped in the hole.
 Drilled out shoe track from 370' to 416'.
 Drilled new formation to 433'.
 Circulated hole clean.
 Performed FIT to a 1 psi/ft. Good test.
 Pulled out of the hole.
 Held detailed safety meeting with all involved personal on picking up BHA.



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Report No: 3

Report For 29-Jun-21

Comments

Test out all BOPE to 300 psi low and 1,000 psi high. All test good.

Casing/Tubular Information

Type	Size (ins)	Top MD (ft)	Top TVD (ft)	Bottom MD (ft)	Bottom TVD (ft)	Hole Section	OH Diam. (ins)	Nom. Wgt. (lbs)	Nominal Grade	LOT (lbs/gal)
FULL	16.000	0		416		SURF	22.000	65	K-55	19.20

Mud Information

%																Gels			Temp		Mud
Dens.	Vis	PV	YP	Filt.	Cake	pH/ES	Solids	Oil	Water	Sand	LGS	Cl	Ca	CaCl	10s	10m	30m	In	Out	Loss	
29-Jun-21 21:00 at Depth 436 ft Mud Pits, Type: Spud Mud																					
8.60	36	8	7	18.8	1	11.5	5	0	95	0.5	0.6	900	240		4	6	11	85	97		

Mud Consumables

Item Description			Qty.	Cost	Item Description			Qty.	Cost
Bicarb - 50#SK			10	---	Desco CF - 25#SK			3	---
Engineering - EACH			1	---	perdiem - OTHER			1	---
wraps - OTHER			2	---					

Bit/BHA/Workstring Information

					Depth	This Run		R.O.P.				Mud		Pump						
No Run		Make	Model	Diam	In	Dist	Hrs	Avg	Max	WOB	RPM	Torque	Wt	Flow	Press	J. Vel	P. Drp	HPH	JIF	
2	1	BAKER	GT-C1	14.750	360	93	1.5	62.0	80.0	10	50	3	9	450	184	194	289	76	388	
Jets: 18 18 18					Out: 433		Grade: Cutter:		1 / 1		Dull: NO / NO		Wear: A		Brgs: 0		Gge:		Pull: BHA	
BHA - No. 3 - BIT, OTHER, STAB, MWD, 4 OTHER, MMTR, RR, FLOAT, 2 OTHER, DC, XO, HWDP = 956.61																				

Mud Log Information

Depth (ft)		TVD (ft)		Gas (Units)			Gas		Drilling Exp.	Pore Press	Mud Dens.	Shale Dens.	ROP	
From	To	From	To	Avg	Max	at Depth	Connect.	Trip					Shale	Sand
422	436													
Formation Name:		Alluvium (sand-cobbles)												
Lithology:		100% Alluvium (granitic)												

Rig Information

Equipment Problems: None.

Location Condition: Good.

Transport:

Solids Control Information

Screen Sizes:		Top	Middle 1	Middle 2	Bottom	Equipment Usage (Hrs):	
Shaker No 1:		60	60	60	60	Desander: 0	Desilter: 0
Shaker No 2:		60	60	60	60		
Shaker No 3:		60	60	60	60		

Bulk Inventory

Item Type	Units	Beginning	Used	Received	On Hand	Net
Rig Fuel		6,697	477		6,220	

Drill Pipe Inventory

DP Size	Joints	Weight	Grade	Thread	DP Size	Joints	Weight	Grade	Thread
5"	114	19.5	S-135	4.5IF					

Safety Information

Meetings/Drills	Time	Description
Safety	60	Two 30 min Pre tour safety meetings with both crews. Covered pinch points, testing, handling BHA
First Aid Treatments: 0 Medical Treatments: 0 Lost Time Incidents: 0 Days Since LTI:		
☒ BOP Test	☒ Crownamatic Check	



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Well ID: Forge 78B-32

Field: UTAHFORGE

Geothermal Resource Group, Inc.

Well Name: 78B-32

Sect: 32 Town: 26 Rng: 9W County: Beaver State: UT

Report No: 3

Report For 29-Jun-21

Weather Information

Sky Condition:	Partly Cloudy	Visibility:	10
Air Temperature:	61 degF	Bar. Pressure:	29.94
Wind Speed/Dir:	4 / NW	Wind Gusts:	1

**Daily Drilling Report**

Well ID: Forge 78B-32

Field: UTAHFORGE

Geothermal Resource Group, Inc.

Well Name: 78B-32

Sect: 32 Town: 26 Rng: 9W County: Beaver State: UT

Report No: 4

Report For 30-Jun-21

Operator: University of Utah		Rig: Frontier Rig 16		Spud Date: 28-Jun-21		Daily Cost / Mud (\$): ---			
Measured Depth (ft): 2699		Last Casing: 16.000 at 416		Wellbore: Original Wellbore		AFE No.		AFE (\$)	Actual (\$)
Vertical Depth (ft): 2699		Next Casing: 11.750 at 3,300		RKB Elevation (ft): 30.40		---		---	---
Proposed TD (ft): 9500		Last BOP Test: 29-Jun-21		Job Reference RKB (ft):		---		---	---
Hole Made (ft) / Hrs: 2,266 / 15.0		Next BOP Test: 19-Jul-21		Working Interest:		Totals:		---	---
Average ROP (ft/hr): 151.07				LOT (lbs/gal): 0.58		Well Cost (\$): ---			
Drilling Days (act./plan): 4/21		Flat Days (act./plan): 0/9		Total Days (act./plan): 4/30		Days On Location: 4			
Pers/Hrs:	Operator: 3 / 36	Contractor:	14 / 168	Service:	4 / 48	Other:	3 / 36	Total:	24 / 288

Safety Summary: No incidents or events reported. Conducted Safety Meeting.

Current Operations: Pulled out of the hole from 951' to Surface.
 Changed bit. Tipped in the hole to 2,699'.
 Drilled 14 3/4" hole section from 2,699' to 2,818' (report time)

Planned Operations: Drill 14 3/4" section to casing point.**Toolpusher:** Steve Caldwell , Justin Bristol**Wellsite Supervisors:** Virgil Welch , Brian Gresham**Tel No.:** 7132807438**Operations Summary**

From	To	Elapsed	End MD(ft)	Code	Operations Description	Non-Prod
0:00	3:30	3.50	433	3-34-3	Picked up 14 3/4" RSS BHA to 433'.	
3:30	4:15	0.75	665	3-2-3	Drilled 14 3/4" hole from 433' to 665' with full returns.	
4:15	4:30	0.25	665	03-051	Circulated hole clean to rack back HWDP and place 3 stands of 8" drill collars into BHA.	
4:30	5:30	1.00	665	3-34-3	Pulled out of the hole from 665' to 389'. Racked back 3 stands of HWDP. Installed 3 stands of 8" drill collars into the BHA. Tripped in the hole to 665'.	
5:30	6:00	0.50	858	3-2-3	Drilled 14 3/4" hole from 665' to 858 with full returns.	
6:00	10:00	4.00	2,077	3-2-3	Drilled 14-3/4" hole from 858' to 2,077' with full returns.	
10:00	10:30	0.50	2,077	SERV	Serviced rig / torqued saver sub	
10:30	11:00	0.50	2,232	3-2-3	Drilled 14-3/4" hole from 2,077' to 2,232' with full returns.	
11:00	11:30	0.50	2,232	REPR	Rig blacked out/ cleaned #2 generator radiator due to overheating.	X
11:30	12:30	1.00	2,359	3-2-3	Drilled 14-3/4" hole from 2,232' to 2,359' with full returns.	
12:30	13:30	1.00	2,359	REPR	Rig Repair cleaned #1 generator radiator due to overheating.	X
13:30	14:00	0.50	2,399	3-2-3	Drilled 14-3/4" hole from 2,359' to 2,399' with full returns.	
14:00	15:30	1.50	2,399	REPR	Rig Repairs cleaned #3 generator radiator due to overheating.	X
15:30	21:15	5.75	2,399	3-2-3	Drilled 14-3/4" hole from 2,399' to 2,699' with full returns.	
21:15	22:00	0.75	2,399	3-5-1	Circulate hole clean. Pump 15 bbl slug.	
22:00	0:00	2.00	2,399	3-6-4	Pull out of the hole from 2,699' to 951'. Tight hole encountered from 2,351' to 1,799' 50-100k over string weight. Extended trip time due to tight connections and using rig tongs. Removed rotating rubber at 850'.	

Management Summary

Picked up 14 3/4" RSS BHA to 433'.
 Drilled 14 3/4" hole from 433' to 665' with full returns.
 Circulated hole clean.
 Pulled out of the hole from 665' to 389'.
 Racked back 3 stands of HWDP.
 Installed 3 stands of 8" drill collars into the BHA.
 Tripped in the hole to 665'.
 Drilled 14 3/4" hole from 665' to 2,077' with full returns. Serviced rig.
 Drilled 14-3/4" hole from 2,077' to 2,232' with full returns. Circulated while cleaning radiator on gen #2 due to overheating.
 Drilled 14-3/4" hole from 2,232' to 2,359' with full returns. Circulated while cleaning radiator on gen #1 due to overheating.
 Drilled 14-3/4" hole from 2,359' to 2,399' with full returns. Circulated while cleaning radiator on gen #3 due to overheating.
 Drilled 14-3/4" hole from 2,399' to 2,699' with full returns. Circulated hole clean. Tripped out of the hole to 951'.

Comments

BOPE test results emailed DWR (Jim Goddard).

Casing/Tubular Information

Type	Size (ins)	Top MD (ft)	Top TVD (ft)	Bottom MD (ft)	Bottom TVD (ft)	Hole Section	OH Diam. (ins)	Nom. Wgt. (lbs)	Nominal Grade	LOT (lbs/gal)
FULL	16.000	0		416		SURF	22.000	65	K-55	0.58

**Daily Drilling Report**

Well ID: Forge 78B-32

Field: UTAHFORGE

Geothermal Resource Group, Inc.

Well Name: 78B-32

Sect: 32 Town: 26 Rng: 9W County: Beaver State: UT

Report No: 4

Report For 30-Jun-21

Mud Information

%																Gels			Temp		Mud
Dens.	Vis	PV	YP	Filt.	Cake	pH/ES	Solids	Oil	Water	Sand	LGS	Cl	Ca	CaCl	10s	10m	30m	In	Out	Loss	
30-Jun-21 21:00 at Depth 2,697 ft Mud Pits, Type: Low Solids Non-Dispersed																					
9.10	42	14	13	9.4	1	10	9	0	91	12.5	4.2	800	120		3	7	11	110	118		

Mud Consumables

Item Description			Qty.	Cost	Item Description			Qty.	Cost
Barite - 100#SK			20	---	Desco CF - 25#SK			1	---
DMA - 50#SK			5	---	Drispac LV - 50#SK			5	---
Engineering - EACH			1	---	FlowZan - 25#SK			10	---
Gel - 100#SK			70	---	Micro C - 50#SK			50	---
pallets - OTHER			13	---	perdiem - OTHER			1	---
Trucking - EACH			1	---					

Bit/BHA/Workstring Information

					Depth	This Run		R.O.P.				Mud		Pump						
No	Run	Make	Model	Diam	In	Dist	Hrs	Avg	Max	WOB	RPM	Torque	Wt	Flow	Press	J. Vel	P. Drp	HP	JIF	
2	1	BAKER	GT-C1	14.750	360	166	3	55.3	1,	40	80	15	9	950	3050	409	1361	754	1829	
Jets: 18 18 18					Out: 433		Grade: Cutter:		1 / 1		Dull: NO / NO		Wear: A		Brgs: 0		Gge:		Pull: BHA	
3	1	NOV	TKC66	14.750	433	2,266	15	151.1					9	950	3050	281	645	357	1259	
Jets: 15 15 15 15 16 16					Out: 2699		Grade: Cutter:		3 / 1		Dull: LT / CT		Wear: N		Brgs: X		Gge:		Pull: PR	
BHA - No. 3 - BIT, OTHER, STAB, MWD, 4 OTHER, MMTR, RR, FLOAT, 2 OTHER, DC, XO, HWDP = 956.61																				

Drilling Parameters

Depth (ft)		ROP (ft/hr)		WOB (lbs)		RPM		Torque (ft lbs)		Flow (gals/min)		Pressure (psi)	
From	To	Avg	Max	Avg	Max	Avg	Max	Avg	Max	Avg	Max		
433	2,699	174.3	1,300.0	40	65	70	120	13	22	950	1,050	3,150	
Annular Velocity: Drill Collars:					197.7	Drill Pipe:					128.2		

Miscellaneous Drilling Parameters

Hook Loads (lbs):		Off Bottom Rotate:		125	Pick Up:		150	Slack Off:		100	Drag Avg/Max:		10 / 15
Slow Circulation Data:													
Pump 1:		35 spm	95 psi	45 spm	210 psi	60 spm	375 psi						
Pump 2:		35 spm	95 psi	45 spm	210 psi	60 spm	375 psi						
Pump 3:		35 spm	95 psi	45 spm	210 psi	60 spm	375 psi						
Hours on BHA:		Since Inspection:		9	Total:		9	Jars:		0			
Hours on Casing/Liner:		Rotating:			9 / 0	Tripping:			3 / 0	☐	Wear Bushing Installed		

Survey Information

Survey Type	Meas. Depth		Inc.	Azimuth	TVD	Closure	Vertical Section	Coordinates		D.L.S.
								N-S	E-W	
MWD	545.0	0.17		93.9	545.0	4.8	4.8	N 4.8	E 0.2	0.513
MWD	731.0	0.07		271.8	731.0	4.8	4.8	N 4.8	E 0.4	0.129
MWD	822.0	0.12		37.4	822.0	4.8	4.8	N 4.8	E 0.4	0.187
MWD	914.0	0.15		342.6	914.0	5.0	5.0	N 5.0	E 0.4	0.138
MWD	1,005.0	0.47		187.3	1,005.0	4.8	4.8	N 4.8	E 0.3	0.670
MWD	1,101.0	0.19		350.9	1,101.0	4.5	4.5	N 4.5	E 0.3	0.682

**Daily Drilling Report**

Well ID: Forge 78B-32

Field: UTAHFORGE

Geothermal Resource Group, Inc.

Well Name: 78B-32

Sect: 32 Town: 26 Rng: 9W County: Beaver State: UT

Report No: 4

Report For 30-Jun-21

Mud Log Information

Depth (ft)		TVD (ft)		Gas (Units)			Gas	Drilling Exp.	Pore Press	Mud Dens.	Shale Dens.	ROP	
From	To	From	To	Avg	Max	at Depth	Connect.					Trip	Shale
0	430	0	430	0	0					9.10			
Formation Name:													
Lithology:		100% Alluvium											
430	2,430	430	2,430	0	0								
Formation Name:													
Lithology:		100% Alluvium											
2,440	2,630	2,430	2,440	0	0								
Formation Name:		60% Clay, 40% Alluvium											
Lithology:		90% Clay 10% Alluvium											
2,630	2,650	2,440	2,570	0	0								
Formation Name:		90% Clay 10% Alluvium											
Lithology:		100% Alluvium											
2,650	2,670	2,570	2,580	0	0								
Formation Name:		70% Clay, 30% Rhyolite											
Lithology:		10-50% Rhyolite 50-90% Alluvium											
2,670	2,700	2,580	2,640	0	0								
Formation Name:		90\$ Rhyolite, 10% Clay											
Lithology:		100% Alluvium											

Rig Information

Equipment Problems: Generators overheating. cleaned radiators on all 3 gens.

Location Condition: Good.

Transport:

Solids Control Information

Screen Sizes: Top Middle 1 Middle 2 Bottom

Shaker No 1:	60	60	60	60
Shaker No 2:	60	60	60	60
Shaker No 3:	60	60	60	60

Bulk Inventory

Item Type	Units	Beginning	Used	Received	On Hand	Net
Rig Fuel		6,220	1,624	3,500	8,096	1,876

Drill Pipe Inventory

DP Size	Joints	Weight	Grade	Thread	DP Size	Joints	Weight	Grade	Thread
5"	114	19.5	S-135	4.5IF					

Safety Information

Meetings/Drills	Time	Description
Safety	60	Two 30 min Pre tour safety meetings with both crews. Trip hazards, pinch points, Working at heights.

First Aid Treatments: 0 Medical Treatments: 0 Lost Time Incidents: 0 Days Since LTI:

☐ BOP Test ☐ Crownamatic Check**Weather Information**

Sky Condition:	Partly Cloudy	Visibility:	10
Air Temperature:	75 degF	Bar. Pressure:	29.94
Wind Speed/Dir:	10 / W	Wind Gusts:	2



Daily Drilling Report

Well ID: Forge 78B-32

Field: UTAHFORGE

Geothermal Resource Group, Inc.

Well Name: 78B-32

Sect: 32 Town: 26 Rng: 9W County: Beaver State: UT

Report No: 5

Report For 01-Jul-21

Operator:	University of Utah	Rig:	Frontier Rig 16	Spud Date:	28-Jun-21	Daily Cost / Mud (\$):	---
Measured Depth (ft):	2699	Last Casing:	16.000 at 416	Wellbore:	Original Wellbore	AFE No.	---
Vertical Depth (ft):	2699	Next Casing:	11.750 at 3,300	RKB Elevation (ft):	30.40	AFE (\$)	Actual (\$)
Proposed TD (ft):	9500	Last BOP Test:	29-Jun-21	Job Reference RKB (ft):	---	---	---
Hole Made (ft) / Hrs:	310 / 22.75	Next BOP Test:	19-Jul-21	Working Interest:	---	Totals:	---
Average ROP (ft/hr):	13.63	LOT (lbs/gal):	0.58	Well Cost (\$):	---	---	---
Drilling Days (act./plan):	5/21	Flat Days (act./plan):	0/9	Total Days (act./plan):	5/30	Days On Location:	5
Pers/Hrs: Operator:	3 / 36	Contractor:	14 / 148	Service:	4 / 48	Other:	3 / 36
				Total:	24 / 268		

Safety Summary: No incidents or events reported. Conducted Crown Check, Safety Meeting.

Current Operations: Cleaned and cleared rig floor. Rigged up and ran 11 3/4" casing to 2,800' (report time)

Planned Operations:

- Run 11 3/4" casing to set depth.
- Rig down Casing crew.
- Run in the hole with stab-in tool.
- Circulate and cement 11 3/4" casing.
- Nipple down 20" BOPE.

Toolpusher: Steve Caldwell , Justin Bristol

Wellsite Supervisors: Virgil Welch , Brian Gresham

Tel No.: 7132807438

Operations Summary

From	To	Elapsed	End MD(ft)	Code	Operations Description	Non-Prod
0:00	2:00	2.00	2,699	3-6-2	Pulled out of the hole from 951' to Surface. Broke out bit and inspected MWD. No visible wear on roller reamer or stabs. Motor had minimal movement in bearing assembly. cleaned and cleared rig floor.	
2:00	4:00	2.00	2,699	3-6-1	Make up 14 3/4" Bit and trip in the hole with BHA to 951'.	
4:00	5:15	1.25	2,699	10-6-3	Trip in the hole from 951' to 2,699'.	
5:15	15:30	10.25	3,009	3-2-3	Drilled 14-3/4" hole from 2,699' to 3,009' with full returns.	
15:30	16:00	0.50	3,009	3-2-3	After making a hook ROP dropped from 15'/hour to 0'/hour. Perform step test from 5K to 30K no ROP / No change in diff pressure and no pressure loss and no hook load change but found excessive amounts of metal on ditch magnet mix dry job.	
16:00	19:30	3.50	3,009	3-6-2	Pulled out of the hole from 3,009' to 951' with no issues. hoe taking proper fill. Stopped and removed rotating rubber at 1,100'.	
19:30	0:00	4.50	3,009	3-6-2	Racked back HWDP. Racked back 8" drill collars. Laid out Cubic sensor, Diffuser sub, along with float sub. Pulled up and broke out bit. laid out the remaining RSS tools. (Note: Bit was 3.75" under gage, 3.5" of the center of the bit was washed out with 4 jets missing. RSS had signs of wear below the pads approximately .25" wide by .25" deep. No signs of wear on the stabilizers or roller reamer. Motor had approximately .25" of play in bearing assembly.	

Management Summary

Pulled out of the hole from 951' to Surface.
Picked up 14 3/4" PDC bit.
Tripped in the hole to 2,699'.
Drilled 14-3/4" hole from 2,699' to 3,009' with full returns. Decision was made to pull out of the hole at 3,009' due to low ROP.
Pulled out of the hole from 3,009' to surface.
Laid out RSS assembly. cleaned and cleared rig floor for casing run.

Comments

Bit was 3.75" under gage with 3.5" of the center of the bit was washed out with 4 jets missing. RSS had signs of wear below the pads approximately 0.25" wide by 0.25" deep. No signs of wear on the stabilizers or roller reamer.

Casing/Tubular Information

Type	Size (ins)	Top MD (ft)	Top TVD (ft)	Bottom MD (ft)	Bottom TVD (ft)	Hole Section	OH Diam. (ins)	Nom. Wgt. (lbs)	Nominal Grade	LOT (lbs/gal)
FULL	16.000	0		416		SURF	22.000	65	K-55	0.58

Mud Information

%														Gels			Temp		Mud Loss
Dens.	Vis	PV	YP	Filt.	Cake	pH/ES	Solids	Oil	Water	Sand	LGS	Cl	Ca	CaCl	10s	10m	30m	In	Out
01-Jul-21 20:00 at Depth 3,009 ft Mud Pits, Type: Low Solids Non-Dispersed																			
9.00	42	14	15	7.8	1	11	9	0	92	1	3.2	800	120		5	9	16	116	109

**Daily Drilling Report**

Well ID: Forge 78B-32

Field: UTAHFORGE

Geothermal Resource Group, Inc.

Well Name: 78B-32

Sect: 32 Town: 26 Rng: 9W County: Beaver State: UT

Report No: 5

Report For 01-Jul-21

Mud Consumables

Item Description	Qty.	Cost	Item Description	Qty.	Cost
Barite - 100#SK	20	---	Engineering - EACH	1	---
Gel - 100#SK	96	---	perdiem - OTHER	1	---
Zanflow - 25#SK	5	---			

Bit/BHA/Workstring Information

				Depth		This Run		R.O.P.				Mud		Pump						
No	Run	Make	Model	Diam	In	Dist	Hrs	Avg	Max	WOB	RPM	Torque	Wt	Flow	Press	J. Vel	P. Drp	HHP	JIF	
3	1	NOV	TKC66	14.750	433	2,266	9	251.8								267	572	301	1118	
Jets: 15 15 15 15 16 16					Out: 2699		Grade: Cutter:		3 / 1		Dull: LT / CT		Wear: N		Brgs: X		Gge:		Pull: PR	
4	1	NOV	TKC63	14.750	2699	310	10.25	30.2	75.0	40	188	15	9	900	2560	267	572	301	1118	
Jets: 15 15 15 15 16 16					Out: 3009		Grade: Cutter:		4 / 8		Dull: RG / CR		Wear: A		Brgs: X		Gge: 3.75		Pull: PR	
BHA - No. 4 - BIT, OTHER, STAB, MWD, 4 OTHER, MMTR, RR, FLOAT, 2 OTHER, DC, XO, HWDP = 956.61																				

Drilling Parameters

Depth (ft)		ROP (ft/hr)		WOB (lbs)		RPM		Torque (ft lbs)		Flow (gals/min)		Pressure (psi)	
From	To	Avg	Max	Avg	Max	Avg	Max	Avg	Max	Avg	Max		
2,699	3,009	30.2	80.0	40	70	188	220	15	21	900	950	2,587	
Annular Velocity: Drill Collars:				199.3	Drill Pipe:				129.3				

Miscellaneous Drilling Parameters

Hook Loads (lbs):		Off Bottom Rotate:		145	Pick Up:		165	Slack Off:		135	Drag Avg/Max:		10 / 15
Slow Circulation Data:													
Pump 1:		30 spm	95 psi	45 spm	210 psi	60 spm	350 psi						
Pump 2:		30 spm	95 psi	45 spm	210 psi	60 spm	350 psi						
Pump 3:		30 spm	95 psi	45 spm	210 psi	60 spm	350 psi						
Hours on BHA:		Since Inspection:		19.25	Total:		19.25	Jars:		0			
Hours on Casing/Liner:		Rotating:		19.25 / 10.25		Tripping:		11 / 8		☐ Wear Bushing Installed			

Mud Log Information

Depth (ft)		TVD (ft)		Gas (Units)		Gas		Drilling		Pore		Mud		Shale		ROP	
From	To	From	To	Avg	Max	at Depth	Connect.	Trip	Exp.	Press	Dens.	Dens.	Dens.	Shale	Sand		
2,650	2,670	2,650	2,670	0	0	0	0	0	0		9.00						
Formation Name: 1050% Rhyolite 50-90% Granite																	
Lithology:																	
2,670	2,700	2,670	2,700	0	0	0	0	0	0		9.00						
Formation Name:																	
Lithology: 100% Granite																	
2,700	3,000	2,700	3,000	0	0	0	0	0	0		9.00						
Formation Name:																	
Lithology: 100% Granite																	

Rig Information

Equipment Problems:	Generators having issues with over heating with higher pump rates.
Location Condition:	Good
Transport:	

Solids Control Information

Screen Sizes:	Top	Middle 1	Middle 2	Bottom	Equipment Usage (Hrs):	
Shaker No 1:	60	60	60	60		
Shaker No 2:	60	60	60	60	Centrifuge 1: 14 (Solids Removal)	Centrifuge 2: 14
Shaker No 3:	60	60	60	60		

Bulk Inventory

Item Type	Units	Beginning	Used	Received	On Hand	Net
Rig Fuel		8,096	2,844		5,252	



Daily Drilling Report

Well ID: Forge 78B-32

Field: UTAHFORGE

Geothermal Resource Group, Inc.

Well Name: 78B-32

Sect: 32 Town: 26 Rng: 9W County: Beaver State: UT

Report No: 5

Report For 01-Jul-21

Drill Pipe Inventory

DP Size	Joints	Weight	Grade	Thread	DP Size	Joints	Weight	Grade	Thread
5"	114	19.5	S-135	4.5IF					

Safety Information

Meetings/Drills	Time	Description
-----------------	------	-------------

Safety 60 Two 30 min Pre tour safety meetings with both crews. Handling BHA, Tripping pipe, Rigging up and running casing.

First Aid Treatments: 0 Medical Treatments: 0 Lost Time Incidents: 0 Days Since LTI:

☐ BOP Test ☒ Crownamatic Check

Weather Information

Sky Condition:	Clear	Visibility:	10
Air Temperature:	70 degF	Bar. Pressure:	29.83
Wind Speed/Dir:	6 / NNW	Wind Gusts:	1